SSD HW19

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(1) (a) $N_D = 10^{16}$ $J_S = 5\times10^{-8}\ A/cm^2$ $N_C = 4.7\times10^{17}/cm^3$
$\varepsilon_r = 13.1$ $A^{**} = 4.4\ A/K^2cm^2$
$J_S = A^{**}T^2 \exp\left(-\frac{\phi_{Bn}}{V_T}\right)$ ϕ_{Bn} → metal semiconductor

$5\times10^{-8} = 4.4\times9\times10^4\ e^{-\phi_{Bn}/0.026}$ $= 0.768V$

$V_{bi} = \phi_{Bn} - V_T \ln\left(\frac{N_C}{N_D}\right)$ V_{bi} → semiconductor -metal
$= 0.668V$

(b) $W = \sqrt{\frac{2\varepsilon V_{bi}}{qN_D}} = 0.311\ \mu m$

(b) $D_p = 10\ cm^2/s$ $\tau_p = 10^{-8}\ s$
$L_p = \sqrt{D_p\tau_p} = 3.16\times10^{-4}\ cm$
$P_{no} = \frac{n_i^2}{N_D} = 4\times10^{-4}/cm^3$

$J_{p_0} = \frac{qD_pP_{no}}{L_p} = 2.03\times10^{-18}\ A/cm^2$

$J_S >> J_{p_0}$

(2) $\beta = 150$ $\alpha_T = 0.995$ $\frac{N_D}{N_P} = 3$ $\frac{W_B}{W_E} = 0.5$
$\alpha = 0.9934$
$\beta = \frac{\alpha}{\alpha_T} = 0.9984$

$\gamma = \frac{1}{1+\left(\frac{D_P}{D_n}\right)\left(\frac{N_{AB}}{N_{DE}}\right)\left(\frac{W_B}{W_E}\right)}$ ($\frac{D_P}{D_n}$: ↓1/3, $\frac{W_B}{W_E}$: ↓1/2) $\Rightarrow \frac{N_{AB}}{N_{DE}} = 0.0098$

(3) $W_{B_1} = 1\mu m$ $W_{B_2} = 2\mu m$
$I_B = \frac{Q_B}{\tau_B}$ $Q_B = \frac{1}{2}\cdot qAW_B\Delta n_{B_0}$

$I_C = qAD_B\frac{\Delta n_{B_0}}{W_B}$ $\Rightarrow I_B = I_C\left(\frac{W_B^2}{2D_B\tau_B}\right)$
$\Rightarrow I_{B_2} = 40\mu A$

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(4) $\beta = 100 \quad V_{BC} + V_{bi} = 1V$

$W_2 = 3W_1 = 0.9\mu m$

$1.2\mu m + 0.3\mu m = 0.9\mu m + W_{B2} \Rightarrow W_{B2} = 0.6\mu m$

a) $\frac{1}{\beta} = \frac{1}{\gamma} - 1 \quad \alpha W_B \Rightarrow \beta \propto \frac{1}{W_B} \Rightarrow \beta_2 = 2\beta_1 = 200$

b) $1 - \alpha_T = \frac{W_B^2}{2L_B^2} \rightarrow (1-\alpha_T) \propto W_B^2 \Rightarrow \alpha_{T2} = \frac{403}{404}$

$\Rightarrow \beta_2 = \frac{\alpha_{T2}}{1-\alpha_{T2}} = 403$

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(1) $V_{th} = 2\phi_B + V_{FB} + \frac{\sqrt{2q\epsilon_{Si} N_A (2\phi_F)}}{C_{ox}} \qquad C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$

t_{ox}, N_A increased $\Rightarrow V_{th} \uparrow$ (d)

(2) $V_{FB} = 0 \qquad V_{th} = 2\phi_B + \frac{\sqrt{2q\epsilon_{Si} N_A (2\phi_F)}\, t_{ox}}{\epsilon_{ox}}$

$\epsilon_{ox} = 3.9\epsilon_0 = 3.453 \times 10^{-13} F/cm$

$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = 3.453 \times 10^{-8} F/cm$

$\phi_F = V_T \ln\left(\frac{N_A}{n_i}\right) = 0.288V$

$Q_d = \sqrt{2\epsilon_{Si} N_A (2\phi_F)} = 1.38 \times 10^{-8} C/cm^2$

$V_{TH} = 2\phi_F + \frac{Q_d}{C_{ox}} = 0.974V$

$N_A = 10^{14} \Rightarrow 0.569V$

$N_A = 10^{16} \rightarrow 2.084V$

$N_A = 10^{17} \Rightarrow 5.564V$

$V_{FB} = \phi_m - \phi_s \qquad \phi_m = 4.1eV$

$\phi_s = \chi_{Si} + \frac{E_g}{2} + \phi_F = 4.61 + \phi_F$

$V_{TH,real} = V_{FB} + V_{TH,ideal}$

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(3) $N = 2 \times 10^{16}$

$\phi_m = 4.1\ eV \quad \chi_s = 4.05\ eV \quad C_{max} = 400pF$

$n_i = 1.5 \times 10^{10}$

a) C_{max} occurs @ -ve V ⇒ -ve voltage on gate attracts holes to oxide-semiconductor interface.

⇒ p-type

b) high

c) $C_{ox} = \frac{C_{max}}{A} = 4 \times 10^{-8}\ F/cm^2$

$t_{ox} = \frac{\epsilon_{ox}}{C_{ox}} = \frac{3.45 \times 10^{-13}}{4 \times 10^{-8}} = 86.3nm$

d) $\phi_F = V_t \ln\left(\frac{N_A}{n_i}\right) = 0.365V$

$\phi_s = \chi_s + \frac{E_g}{2} + \phi_F = 4.05 + 0.56 + 0.365 = 4.975$

$V_{FB} = \phi_m - \phi_s = -0.875V$

e) $V_{th} = V_{FB} + 2\phi_F + \frac{Q_{d,max}}{C_{ox}} \qquad Q_{dmax} = \sqrt{2q\epsilon_{Si} N_A (2\phi_F)}$

$= 1.595V$

f) C_{min} occurs when W (depletion width) is max

$W_{max} = \sqrt{\frac{2\epsilon_{Si}(2\phi_F)}{qN_A}} = 2.17 \times 10^{-5}\ cm$

$C_d' = \frac{\epsilon_{Si}}{W_{max}} = 4.77 \times 10^{-8}\ F/cm^2$

$C_{min} = \frac{C_{ox} C_d}{C_{ox} + C_d} = 217.5pF$

(4) $I_D = \frac{\mu_n C_{ox} W}{L}\left[(V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2}\right]$

$\Delta V_{TH} = \gamma\left(\sqrt{|2\phi_B - V_S|} - \sqrt{|2\phi_B|}\right)$

$V_{TH} = 1V \quad V_{GS} = 2.1V \quad V_{DS} = 1V \quad 2\phi_B = 0.9V$